

FOCUSED UPDATES

Content Advisors: Giuseppe Lanzino, MD, and Johanna Ospel, MD, PhD

Inflammation in Acute Ischemic Stroke: Translational Barriers and Implications for Treatment

Alexandre Dias¹, MSc; Ana Moutinho², MSc; Liliana Igreja³, MD; João Moura⁴, MD; Susanne Wegener⁵, MD, PhD; Luís F. Maia⁶, MD, PhD

ABSTRACT: Cerebral ischemia triggers a cascade of molecular events over minutes to days, which activate immune cells, inflammation, and coagulation, promoting cell death. These processes critically influence both short- and long-term outcomes in patients with stroke. In this review, we address the role of immune molecular and cellular factors, and the therapeutic approaches predicted to be targetable by specific agents under a timely application. We will focus on translational efforts targeting (thrombo)-inflammatory mechanisms, highlighting past missteps and current opportunities. Finally, we outline key mechanistic and methodological questions that may define a path forward.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: blood-brain barrier ■ clinical trial ■ disease models, animal ■ inflammation ■ ischemic stroke ■ thromboinflammation

Over the last 3 decades, reperfusion treatments have substantially improved acute ischemic stroke (AIS) care. Since 2015, when endovascular treatment (EVT) became the first-line treatment for AIS due to large vessel occlusion, the number of patients benefiting from EVT increased worldwide, and its indications kept expanding. Yet, only 15% to 25% of patients are eligible for reperfusion treatments.¹ Despite up to 90% of cases achieving successful recanalization, <50% of EVT-treated patients reach functional independence, a phenomenon referred to as futile recanalization.²

Inflammatory and immune-mediated responses triggered during AIS are likely responsible for futile recanalization in many instances, and they represent one of the most compelling therapeutic targets to rescue brain tissue.

In this review, we synthesize current knowledge of innate and adaptive immune responses in AIS, examine previous translational missteps, and dissect emerging therapeutic strategies designed to act in synchrony with recanalization, considering the complex interplay between time, cellular or molecular target, and physiological compartment. These

insights define new directions and preclinical advances needed for next-generation stroke therapies.

ISCHEMIA PATHOPHYSIOLOGICAL CASCADE AND NEUROINFLAMMATION

Cerebral ischemia abruptly interrupts oxygen and glucose delivery to the brain parenchyma, leading to rapid ATP depletion, failure of energy-dependent ion pumps, membrane depolarization, ionic imbalance, and glutamate-mediated excitotoxicity. These events trigger oxidative and nitrosative stress, mitochondrial dysfunction, and cell death in the ischemic core within minutes of onset.^{3,4} Damage to neurons, glial cells, and vascular elements compromises the neurovascular unit and blood-brain barrier (BBB) integrity, promoting the release of reactive oxygen species, cytokines, chemokines, and danger-associated molecular patterns, including HMGB1 (high mobility group box 1) and S100 (calcium-binding protein family) proteins. These signals activate innate immune receptors, such as TLRs (toll-like receptors) and NOD (nucleotide-binding oligomerization

Correspondence to: Luís F. Maia, MD, PhD, Department of Neurology, Unidade Local de Saúde de Santo António, ULSSA, Largo Prof. Abel Salazar, 4099-001 Porto, Portugal. Email luismaia.neurologia@ulssa.min-saude.pt

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.126.056147>.

© 2026 American Heart Association, Inc.

Stroke is available at www.ahajournals.org/journal/str

domain)-like receptors on resident immune cells.⁵ Within minutes to hours, microglia and other neurovascular cells, such as astrocytes, become activated, amplifying endothelial dysfunction and upregulating adhesion molecules (eg, VCAM-1 [vascular cell adhesion molecular 1], ICAM-1 [intercellular adhesion molecule 1], and selectins), thereby facilitating infiltration of circulating leukocytes and propagation of the neuroimmune response, ultimately contributing to secondary injury and penumbral expansion (Figure 1).^{6–8}

Systemic Inflammatory Cascade

The systemic inflammatory cascade involves a massive increase in circulating proinflammatory mediators (IL [interleukin]-6, TNF- α [tumor necrosis factor- α], and IL-1 β) in both humans and large vessel occlusion animal models.⁸ Early cytokine release is associated with neutrophil degranulation, exacerbating systemic inflammation during the hyperacute phase.⁹ Leukocyte mobilization from the bone marrow, spleen, and lymph nodes further contributes to this response.^{10–12} Over the following hours to days, there is a shift toward immunosuppression. Hypothalamic-pituitary-adrenal axis activation releases cortisol, impairing lymphopoiesis in the bone marrow, and induces apoptosis of circulating lymphocytes. Regulatory T cells also have a significant role in shifting the landscape toward immunosuppression. CD (cluster of differentiation)⁸⁺T cells rise after the acute phase, exceeding baseline by 3 months. Higher levels during recovery correlate with better prognosis and smaller infarcts.^{5,8,13} However, clearly distinguishing the initial proinflammatory phase from the later immunosuppressive phase is difficult, as they may overlap. Figure 1 illustrates dynamic inflammatory changes in central and peripheral compartments.

Intracranial Inflammatory Cascade

With the advent of EVT for large vessel occlusion treatment, paired blood sampling from systemic, carotid, and intracranial sites has revealed that leukocyte activation and accumulation, particularly neutrophils, peak within the intracranial circulation.^{14,15} These cells contribute to impaired collateral flow and poor outcomes in humans and in a preclinical model.^{15,16} While platelet counts are not consistently elevated intracranially, local platelet activation is suggested by increased chemokine levels, highlighting platelet-leukocyte interactions as drivers of early vascular dysfunction.^{14,15} The role of primary thrombus composition and origin (AIS cause) in the intracranial inflammatory cascade and microvasculature occlusion is largely unknown (Figure S1).

Recanalization, Reperfusion, and Thromboinflammation

While recanalization restores macrovascular flow, adequate microvascular perfusion depends on the integrity

of the neurovascular unit and the absence of microvascular obstruction.² Futile recanalization can occur when recanalization through thrombolysis or EVT does not translate into effective tissue-level reperfusion.² Several studies suggest that earlier and more complete recanalization is associated with reduced systemic inflammatory biomarkers, reduced edema formation, and more stable BBB integrity.¹⁷ Thrombectomy, thus, acts not only as a mechanical intervention but also as a biological modifier that can blunt the trajectory of inflammation if performed within the optimal time window. EVT technique also matters, as prolonged procedures and multiple passes increase endothelial injury and inflammation.^{17,18} The choice of technique between aspiration, stent retriever, and a combined approach also modulates the extent of endothelial trauma and embolic risk, while balloon guide catheters can reduce distal embolization and procedural time, indirectly limiting inflammatory activation.¹⁹

The discrepancy between a successful macrovascular recanalization and unfavorable outcomes is largely attributable to 2 interrelated mechanisms: the ischemia-reperfusion injury and failure of microvascular reperfusion (no-reflow). Both processes are driven by a synergistic interplay between thrombosis and inflammation pathways, collectively referred to as thromboinflammation.²⁰

Thromboinflammation partially sustains both microvascular obstruction and ischemia-reperfusion injury, with each process amplifying the other and contributing to infarct expansion (Figure 2).² Elucidating the complex immune and thromboinflammatory cascades that occur during stroke is essential for developing effective therapeutic strategies, and experimental models constitute a critical platform for this purpose.

MODELS USED IN PRECLINICAL AIS RESEARCH

Despite a deep understanding of AIS mechanisms and numerous preclinical and clinical studies showing promising results for >1000 candidate neuroprotective stroke drugs and nearly 200 clinical trials, no effective therapy other than tissue-type plasminogen activator has been approved.²¹ This discrepancy partly reflects the limited ability of *in vitro* and *in vivo* models to mimic IS complexity.

In Vitro Models

In vitro systems represent the first tier of mechanistic stroke research, ideal to evaluate responses of specific cell types to ischemia with fewer ethical issues than animal models.²¹ The most physiological and common approach to model IS *in vitro* is oxygen-glucose deprivation (OGD).²²

Two-dimensional cell monocultures, primarily rodent primary neurons or immortalized cell lines (eg, SH-SY5Y), are the main *in vitro* platform.²² Primary neurons,

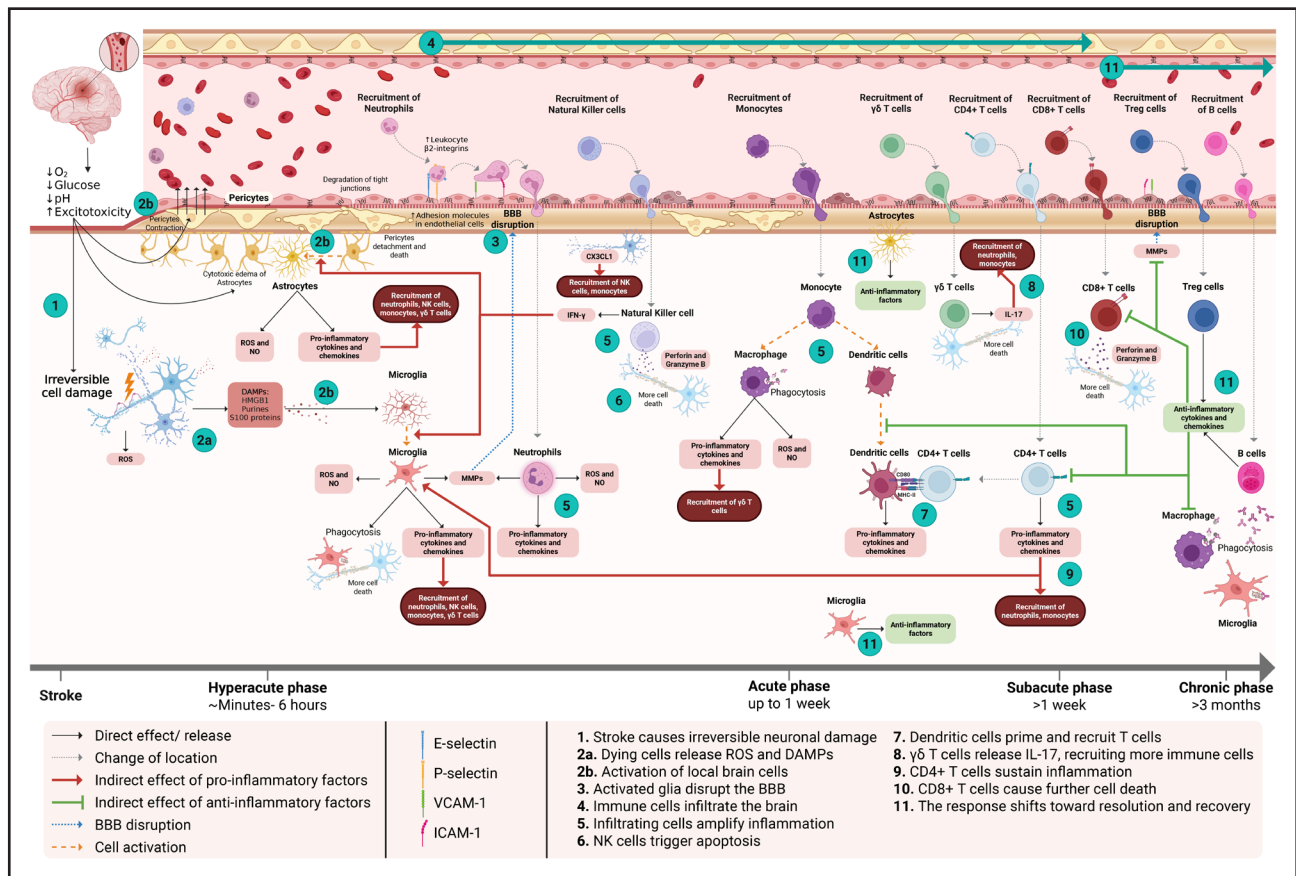


Figure 1. Spatiotemporal inflammatory responses after acute ischemic stroke.

Following a stroke, the decrease in oxygen and glucose and increased excitotoxicity lead to cellular damage (1). Damaged cells release reactive oxygen species (ROS) and danger-associated molecular patterns (DAMPs), such as HMGB1 (high mobility group box 1), purines, and S100 (calcium-binding protein family) proteins (2a). DAMPs activate pericytes, endothelial cells, microglia, and astrocytes, initiating the inflammatory response (2b). Activated microglia and astrocytes secrete proinflammatory cytokines and chemokines (TNF [tumor necrosis factor]- α , IL [interleukin]-1 β , IL-6, IL-15, and CCL [C-C motif chemokine ligand] 2), promoting blood-brain barrier (BBB) disruption by releasing MMPs that degrade tight junctions and by upregulating the expression of adhesion molecules in endothelial cells (E-selectin, P-selectin, VCAM-1 [vascular cell adhesion molecule 1], and ICAM-1 [intercellular adhesion molecule 1]; 3). These changes allow peripheral immune cells, such as neutrophils, natural killer (NK) cells, monocytes, and T lymphocytes, to infiltrate the brain (4). Infiltrating immune cells perpetuate inflammation by additionally producing ROS, NO, MMPs (matrix metalloproteinases), and proinflammatory cytokines and chemokines (5). NK cells release perforin and granzyme B, further triggering apoptosis (6). Inside the brain parenchyma, monocytes can differentiate into dendritic cells that act as antigen-presenting cells and further exacerbate inflammation by producing cytokines and recruiting $\gamma\delta$ T cells (7). $\gamma\delta$ T (gamma delta T) cells are the major source of IL-17, a powerful proinflammatory cytokine that further recruits immune cells (8). CD (cluster of differentiation) 4+ T cells secrete more cytokines (9), while CD8+ T cells secrete perforin and granzyme B, leading to more cell death (10). As inflammation progresses, numerous cells shift to an anti-inflammatory phenotype, and regulatory T cells and B cells start to infiltrate the parenchyma, increasing secretion of anti-inflammatory and neuroprotective mediators, such as IL-10, TGF- β (transforming growth factor β), and IGF1 (insulin-like growth factor 1), counterbalancing injury progression by inhibiting T cells, macrophages, MMPs, and dendritic cells (11). Created in BioRender. Soares, P. (2026) <https://BioRender.com/5iw6eyf>. BDNF indicates brain-derived neurotrophic factor; CXCL, C-X-C motif chemokine ligand; IFN- γ , interferon γ ; and MHC, major histocompatibility complex.

astrocytes, microglia, and endothelial cells recapitulate part of the metabolic and inflammatory responses observed in ischemic tissue, including reactive oxygen species production, danger-associated molecular patterns release, and downstream cytokine secretion.^{21,23,24} However, the absence of vascularization and tissue architecture limits their ability to model cell-cell interaction.²²

Brain or organotypic slices partially overcome these limitations, preserving viable tissue structure and synapse circuitry that enables focal OGD induction.²² Nevertheless, their short viability (around 12 hours for rodent

brain slices) and limited availability of human samples make this model impractical for high-throughput and mechanistic research.^{21,22}

Three-dimensional cerebral organoids derived from human pluripotent stem cells represent a more recent advance, recapitulating cortical cytoarchitecture and human-specific transcriptomic signatures that are absent in both 2-dimensional and rodent-derived cultures.²⁵ Ischemic injury through OGD leads to quantifiable neuronal death (eg, secreted neurofilament light chain) and enables pharmacological screening in a human-tissue

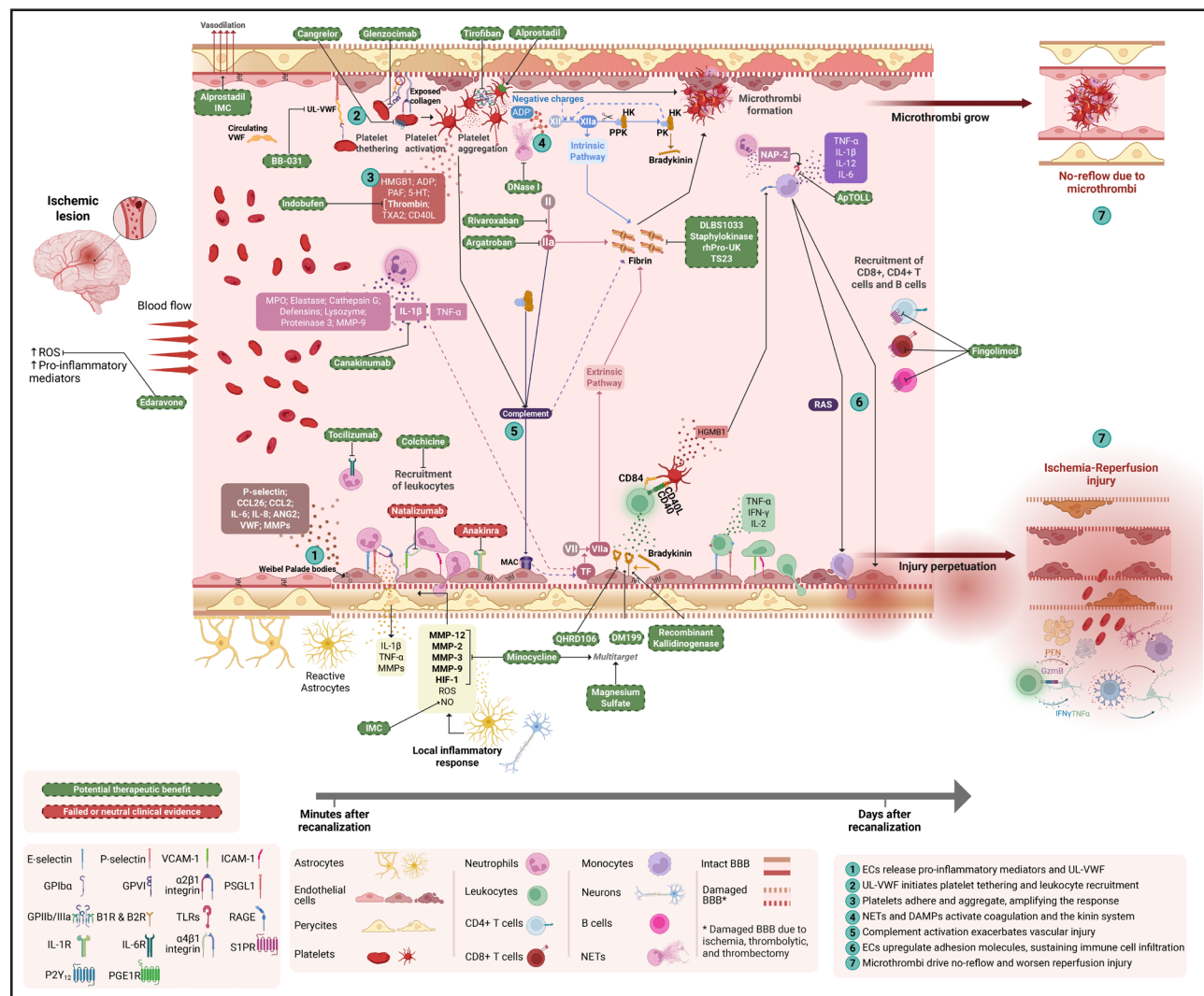


Figure 2. Therapeutic modulation of early thromboinflammatory events in ischemic stroke.

Reperfusion injury triggers thromboinflammation driven by endothelial cells (ECs), platelets, and innate immune cells. Therapeutic strategies aim to attenuate these events by targeting distinct yet interconnected molecular pathways. Mechanistically, ECs release Weibel-Palade bodies containing UL-VWF (ultralarge von Willebrand Factor), FVIII (coagulation factor VIII), and proinflammatory mediators (eg, P-selectin, IL [interleukin]-6, IL-8, and MCP-1 [monocyte chemoattractant protein-1]; 1). Under shear stress, UL-VWF binds platelet GPIIb/IIIa (glycoprotein IIb/IIIa), initiating platelet tethering, while simultaneously recruiting neutrophils and monocytes via PSGL-1 (P-selectin glycoprotein ligand-1) and β_2 integrins (2). Platelets adhere to exposed subendothelial collagen (via GPIIb/IIIa) and integrin $\alpha_2\beta_1$ and fibrin, leading to integrin $\alpha_{IIb}\beta_3$ (GPIIb/IIIa) activation and aggregation, further amplified by HMGB1 (high mobility group box 1), ADP, thrombin, and thromboxane A_2 (3). Activated neutrophils release neutrophil extracellular traps (NETs), entrapping red blood cells (RBCs), platelets, and fibrin, stabilizing microthrombi. Concurrently, polyphosphate, NET-derived DNA, and collagen activate the intrinsic coagulation cascade via FXII (coagulation factor XII) and the contact-kinin system, producing bradykinin and activating B1R (bradykinin receptor B1)/B2R (bradykinin receptor B2) signaling, compromising blood-brain barrier (BBB) integrity (4). Platelet and neutrophil-derived mediators also trigger complement activation, leading to P-selectin upregulation and MAC (membrane attack complex) formation, exacerbating vascular leakage and neuronal injury (5). Within minutes to hours, ECs upregulate VCAM-1 (vascular cell adhesion molecule 1) and ICAM-1 (intercellular adhesion molecule 1), facilitating neutrophil, monocyte, and T-cell infiltration (6). This cascade sustains microthrombus formation within the cerebral microvasculature, driving the no-reflow phenomenon and simultaneously exacerbating ischemia-reperfusion injury (7). Created in BioRender. Soares, P. (2026) <https://BioRender.com/vwlgon4>. 5-HT indicates serotonin; CCL, C-C motif chemokine ligand; CD, cluster of differentiation; CD40L, CD40 ligand; DAMP, damage-associated molecular pattern; HIF-1, hypoxia-inducible factor-1; HK, high-molecular-weight kininogen; IFN- γ , interferon- γ ; IMC, isosorbide mononitrate and cilostazol; MMP, matrix metalloproteinase; MPO, myeloperoxidase; NAP-2, neutrophil-activating peptide 2; PAF, platelet-activating factor; PK, plasma kallikrein; polyP, polyphosphate; PPK, plasma prekallikrein; RAGE, receptor for advanced glycation end products; RAS, renin-angiotensin system; ROS, reactive oxygen species; TLR, toll-like receptor; TNF- α , tumor necrosis factor α ; TXA2, thromboxane A_2 ; and VWF, von Willebrand Factor.

context.²⁶ However, organoids are often heterogeneous within batches and lack mesodermally derived microglia and circulating leukocytes, and the absence of functional

vasculature creates a hypoxic necrotic core that limits its longevity and may confound ischemic interpretation.^{26–28} Consequently, while organoids are a valuable

human-specific platform for dissecting neuronal injury mechanisms and screening for neuroprotective candidates, they still cannot model central biological pathways necessary to study absorption, distribution, metabolism, and excretion of pharmaceutical compounds.²⁸

Microfluidic organ-on-a-chip devices directly address the vasculature gap. Neurovascular unit on-chip systems integrate perfused microchannels, hemodynamic shear stress, and spatially compartmentalized cocultures of endothelial cells, pericytes, astrocytes, neurons, and microglia, and can model ischemia-reperfusion injury through combined OGD and halted perfusion.²⁹ These platforms recapitulate key ischemic signaling events, including endothelial activation and barrier breakdown, which are largely inaccessible in static culture models.²⁹ Limitations include nonstandardized fabrication, limited throughput, and inability to recapitulate systemic immune mobilization. Nevertheless, emerging platforms integrating vascularized organoids within microfluidic circuits may substantially close the species-translation gap for both mechanistic research and drug screening.³⁰

Despite their value for mechanistic studies of cellular responses, *in vitro* models lack key *in vivo* stroke features, including cerebral blood flow, shear stress, thrombus formation, leukocyte trafficking, and systemic immune interactions. They also cannot replicate the mechanical and biological effects of distal embolization or microvascular obstruction characteristic of the postthrombectomy environment. Thus, their translational relevance depends on subsequent validation in *in vivo* models.

Animal Models

Animal models capture vascular and thromboinflammatory complexity, yet key limitations hinder clinical translation. Rodent models still account for ≈90% of studies investigating therapeutic strategies in ischemic stroke and are still increasing in recent years (Figure S2), despite longstanding recommendations to incorporate large animal models (LAMs), such as nonhuman primates (NHPs), which better recapitulate human physiology.³¹

Small Rodent Models

The frequent use of rodent models, such as mice and rats, reflects their practical advantages, including lower ethical barriers, reduced acquisition and housing costs compared with LAMs, rapid reproduction, and reproducible ischemic stroke.³² Key pathophysiological mechanisms are conserved between rodents and humans, such as TLR-mediated innate immune signaling and neutrophil-endothelial interactions via ICAM-1 and selectins, validating the translational relevance of these models. In addition, rodents allow precise genetic manipulation that can be integrated with omics and advanced imaging techniques, at an impractical scale compared with LAMs.

Collectively, rodents have been fundamental for elucidating the molecular and cellular mechanisms underlying ischemic injury and neuroinflammation, and for proposing therapeutic strategies subsequently evaluated in clinical trials. For instance, intravital 2-photon imaging in mice showed that ischemia reperfusion rapidly activates neutrophils, which obstruct brain capillaries, promoting no-reflow, thereby identifying leukocyte-endothelial interactions as a potential therapeutic target.¹⁶

The method of stroke induction also significantly shapes the ischemic cascade, with different models capturing distinct pathophysiological facets (reviewed in the study by Sommer,³³ Singh et al,³⁴ Dias et al,³⁵ and Fluri et al³⁶). Transient filament middle cerebral artery occlusion (MCAO) allows controlled reperfusion, providing a controlled environment for studying adjunctive EVT strategies; embolic models enable spontaneous or thrombolysis-mediated recanalization; and permanent occlusion models lack reperfusion entirely. Photothrombotic approaches offer precise infarct targeting necessary for mechanistic studies, but they lack the significant penumbral dynamics seen in human stroke and generate distinct immune/inflammatory responses.³⁷

Nevertheless, important biological and experimental limitations constrain direct translation. Rodents have smaller, lissencephalic brains with lower white matter composition, and a simpler vascular network and architecture compared with humans.³² Epigenomic, transcriptomic, and proteomic studies have revealed limited overlap between rodent and human immune profiles, including a lymphocyte-dominant profile in rodents versus the neutrophil-predominant profile in humans, and substantial differences in chemokine/cytokine secretion by primary microglial, neuronal, and astrocytic cultures in response to OGD across species.^{33,38,39}

Moreover, many studies still rely on young, healthy animals, which fail to capture the multimorbidity of human patients with stroke. Comorbidities such as hypertension, diabetes, hyperlipidemia, and atherosclerosis profoundly impact infarct progression, collateral dynamics, inflammatory responses, and therapeutic efficacy, further limiting direct translation. While no animal model can recapitulate human complexity and heterogeneity, rodent models have been indispensable for dissecting the mechanisms of ischemia. Indeed, many key concepts in stroke biology, including the ischemic penumbra, early and late inflammatory responses, BBB dysfunction, ischemia-reperfusion mechanisms, and microvascular failure, were first characterized in rodents.

Large Animal Models

The need for experimental systems approximating human anatomic and hemodynamic complexity has become increasingly evident. LAMs offer several advantages, including comparable anatomy and physiology to humans. These

species generally possess gyrencephalic cortex, higher white matter content (dog: 35%; rabbits: 20%; swine: 28%; sheep: 28%; and NHPs: 28%) with similar distribution and organization, and an identical cerebral vasculature, particularly in the middle cerebral artery territory.^{32,33} LAMs also allow the use of clinical diagnostic and therapeutic tools identical to those used in patients, enabling the testing of novel therapeutic strategies and devices, and their larger circulating blood volume facilitates repeated sample collection and longitudinal analysis for studying pharmacodynamic and blood biomarker kinetics.³²

Nevertheless, these models present disadvantages, including higher costs, complex ethical issues, and the requirement of specialized personnel. Functional and behavioral assessments are challenging, often requiring pretraining and prolonged recovery.³² In addition, immunologic differences remain: leukocyte distribution differs from humans, with NHPs having the most similar profile.^{40,41} However, immune cell subpopulations still vary, as NHPs exhibit significantly higher abundance of CD8⁺, CD4⁺, and CD4⁺/CD8⁺ T cells and roughly 3× the number of B cells compared with humans.⁴² The expression of surface receptors at the same class of immune cells also differs across species, likely leading to distinct inflammatory responses following AIS.

In both rodent and LAMs, stroke must be experimentally induced, typically under anesthesia or using artificial clots. These conditions alter systemic and neuroinflammatory responses compared with spontaneous human stroke, introducing bias into preclinical findings.⁴³ In fact, common anesthetics (eg, isoflurane and ketamine/xylazine) have intrinsic neuroprotective properties by modulating physiological processes, such as blood pressure, cerebral blood flow, and metabolism, which may confound results.⁴³ Therefore, awake models should be considered when feasible.⁴⁴

Such interspecies discrepancies underscore the complexity of modeling stroke neuroinflammation and highlight the importance of careful model selection and interpretation of both mechanistic and therapeutic preclinical findings.

THROMBOINFLAMMATION-TARGETED THERAPIES IN STROKE: BARRIERS TO SUCCESSFUL TRANSLATION

Over decades, several studies and trials aimed to improve AIS care by interfering with distinct pathophysiological pathways using anti-inflammatory and immunomodulators.⁴⁵ Their failures reflect the inherent complexity of the stroke-associated inflammatory cascade (Figure 1), the influence of timing, recanalization status, the physiological compartment targeted, and the comorbidities of patients with AIS. Such complexity suggests that a one-size-fits-all approach is unlikely to succeed.

Lessons from past trials showed that, despite robust preclinical efficacy, all major anti-inflammatory and anti-thrombotic agents evaluated in the pre-EVT era failed to demonstrate clinical benefit and, in several cases, caused harm. Table 1 summarizes 5 landmark trials with their preclinical context and the reasons for translational failure now identifiable in retrospect. Recurrent arguments include the use of young, comorbidity-free rodent models, an absent or poorly characterized reperfusion context, xenogeneic or species-incompatible pharmacology, and inadequate modeling of hemorrhagic risk.

Therapies usually show strong efficacy in young, healthy animals treated within minutes, rarely translating to older, comorbid patients treated hours after stroke onset. Aging profoundly alters the cerebral vasculature, impairing reparative mechanisms such as neurogenesis and angiogenesis while sustaining chronic neuroinflammation, impacting infarct size.⁴⁶ Common comorbidities such as hypertension, diabetes, obesity, and hyperlipidemia further exacerbate endothelial dysfunction and BBB fragility. Accordingly, aged and comorbid rodent models consistently exhibit exacerbated BBB damage, cerebral edema, increased mortality, and worse functional recovery.⁴⁶ Yet, <15% of preclinical studies include aged or comorbid models due to higher costs, increased mortality, extended experimental duration, and greater variability in infarct volumes.⁴⁷ Inadequate modeling of reperfusion, drug interactions, and hemorrhagic risks further contributed to clinical futility or harm.

Critically, these failures should not be interpreted as evidence that preclinical research was without value and, rather, reflect a systematic mismatch between the experimental conditions under which efficacy was demonstrated and the biological reality of patients with AIS.

LEARNING FROM FAILURE: TOWARD CLINICAL TRANSLATION

The introduction of EVT has provided new insights into the pathophysiological processes occurring in the intracranial compartment and opened perspectives in precision interventions targeting the thromboinflammatory cascade (Figure 2). Emerging strategies reflect a translational shift from single-mechanism neuroprotection to multifactorial modulation of the inflammatory, thrombotic, and vascular microenvironment in AIS. Table 2 summarizes key trials examined below, which have shown positive or still unknown results with strong preclinical evidence from stroke models, including only ongoing or completed trials post-2015 (after EVT became standard of care). A comprehensive compilation of trials with drugs targeting thromboinflammation and coagulation pathways is available in Table S1. Trials are ordered according to the processes targeted:

Table 1. Summary of Failed Landmark Clinical Trials on Inflammation-Related Targets

Trial	Phase	Size	Start	Target	Therapeutic agent	Mechanism	Timing	Reperfusion	Preclinical model	Preclinical rationale	Key results	Key translational challenges identified
Enlimomab Acute Stroke Trial	III	625	1994	ICAM-1	Enlimomab	Murine anti-ICAM-1 mAb; inhibits leukocyte adhesion/transmigration	Hyper-acute/acute	None	Rodent (tMCAO); rabbit embolic	Decreased infarct size; improved neurological outcomes	Increased mortality; worse neurological outcomes; increased fever/infection	Xenogeneic immunogenicity unpredicted; no reperfusion context
ASTIN	II	966	2000	CD11b/CD18	UK-279,276	Recombinant neutrophil inhibitory factor; binds CD11b/CD18; inhibits neutrophil adhesion/transmigration	Hyper-acute/acute	±IVT (mostly monotherapy)	Rodent (tMCAO)	Reduced neutrophil infiltration and infarct size	No improvement at 90 d	Untested IVT interaction; pharmacokinetic/pharmacodynamic uncertainty; modest preclinical effect size
ESS (NCT00604630)	II/III	522	2003	Erythropoietin receptor	Erythropoietin	Erythropoietin receptor agonist; antiapoptotic, anti-inflammatory, neurotrophic, BBB-stabilizing	Hyper-acute/acute	±IVT	Rodent (tMCAO, in vitro OGD)	Reduced infarct size; improved neurological outcomes (rodent); decreased apoptosis (in vitro)	No benefit; increased mortality; increased sICH/thromboembolism (especially with IVT)	IVT interaction unpredicted; timing mismatch between preclinical/clinical settings; no aged/comorbid models
AbESTT-II (NCT00073372)	III	808	2003	GP1Ib/IIla	Abciximab	Anti-GP1Ib/IIla mAb; reduces platelet aggregation/thromboinflammation	Hyper-acute/acute	None	NHP intracranial thrombosis	Decreased microthrombosis	No benefit; markedly increased sICH; early trial termination	Hemorrhagic risk underestimated preclinically; comorbidities underrepresented in models; reperfusion interaction unclear
ALIAS (NCT00235495)	III	1275	2006	Pleiotropic	Human serum albumin (high-dose)	Volume expansion; anti-inflammatory/antioxidant; increases microvascular perfusion	Hyper-acute/acute	±IVT±EVT	Rodent (tMCAO)	Reduced infarct/edema; increased microvascular flow; improved behavior	No benefit; increased pulmonary edema, congestive heart failure, and ICH	Volume overload in elderly/comorbid patients is underestimated; cardiac comorbidities are underrepresented preclinically; fluid/reperfusion interactions are unmodeled

AbESTT-II indicates Abciximab in Emergency Stroke Treatment Trial II; ALIAS, Albumin in Acute Stroke; ASTIN, Acute Stroke Therapy by Inhibition of Neutrophils; BBB, blood-brain barrier; CD, cluster of differentiation; ESS, Erythropoietin Stroke Study; EVT, endovascular treatment; GP1Ib/IIla, glycoprotein 1Ib/IIla; ICAM-1, intercellular adhesion molecule 1; ICH, intracranial hemorrhage; IVT, intravenous thrombolysis; mAb, monoclonal antibody; NHP, nonhuman primate; OGD, oxygen-glucose deprivation; sICH, symptomatic intracranial hemorrhage; and tMCAO, transient middle cerebral artery occlusion.

Table 2. Key Clinical Trials Post-2015 Targeting Inflammation in Stroke

Key trials	Status	Acronym	Phase	Estimated size	Start date	Target	Drug	Recana- lization treatment	Mechanism	Timing	Key results
Phase 2–phase 4											
NCT06107725	Recruiting	N/A	Phase 2 Phase 3	1120	2023	MMPs and microglia	Minocycline	Yes (STC)	Inhibits MMP-9; modulates HIF-1 α ; suppresses microglial activation	Acute	N/A
NCT05836740	Completed	EMPHASIS	Phase 3	1724	2023			Yes (STC)		Acute	Improved 90-d functional outcome; no significant safety differences
NCT05512910	Not recruiting	MIST-B	Phase 4	90	2023			Yes (EVT)		Hyper-acute/acute	N/A
NCT05487417	Recruiting	MIST-A	Phase 4	180	2022			Yes (EVT)		Acute	N/A
NCT03357133	Completed	MR TEA	Phase 2 Phase 3	30	2017	GPIIb/IIIa	Tirofiban	Yes (IVT)	Blocks GPIIb/IIIa; inhibits platelet aggregation	Hyper-acute	No increase in sICH or mortality; apparent neurological benefit
NCT05728333	Recruiting	ADJUVANT	Phase 2 Phase 3	800	2023			Yes (EVT)		N/A	
NCT05604638	Not recruiting	ATTIS	Phase 4	600	2023			Yes (IVT)		N/A	
NCT02562183	Completed	RESK	Phase 4	2186	2015	Low-MW kininogen	Recombinant urinary-kallidinogenase	No	Activates kallikrein-kinin system; enhances microvascular perfusion	Acute/subacute	Safe and tolerable; improved neurological and functional outcomes
NCT04102956	Completed	N/A	Phase 4	80	2017			No		Hyper-acute/acute/subacute	Improved motor recovery; possible corticospinal-tract protection
NCT06848894	Recruiting	UNITE	N/A	1204	2025			Yes (IVT/EVT)		Hyper-acute	
Phase 2 to phase 3											
NCT05070260	Completed	ACTISAVE	Phase 2 Phase 3	438	2021	Glycoprotein VI	Glenzocimab	Yes (IVT $\pm$ EVT)	Blocks GPVI; reduces platelet-collagen adhesion	Hyper-acute	N/A
NCT05559398	Not recruiting	GREEN	Phase 2 Phase 3	260	2022			Yes (EVT $\pm$ IVT)			N/A
NCT06437431	Not recruiting	GALICE	Phase 2 Phase 3	304	2026			Yes (EVT $\pm$ IVT)			N/A
NCT06238024	Completed	IRIS	Phase 2 Phase 3	108	2024	Interleukin-6 receptor	Tocilizumab	Yes (EVT)	Inhibits interleukin-6 receptor	Hyper-acute	Reduced infarct core volume; potential functional benefit; good safety
NCT05065216	Recruiting	ReMEDy2	Phase 2 Phase 3	728	2021	Tissue kallikrein	DM199 (rinvecalinase alfa)	No	Activates tissue-kallikrein pathway	Hyper-acute/acute/subacute	N/A
NCT04950920	Completed	TASTE-SL	Phase 3	914	2021	Reactive oxygen species (mainly)	Sublingual edaravone dextrobenol	No	ROS modulation	Hyper-acute/acute/subacute	Improved 90-d outcomes
NCT04667637	Completed	INSIST-ED	Phase 2	200	2021		Intravenous edaravone dextrobenol	Yes (EVT)		Hyper-acute/acute/subacute	Safe; no functional benefit

(Continued)

Table 2. Continued

Key trials	Status	Acronym	Phase	Estimated size	Start date	Target	Drug	Recana- lization treatment	Mechanism	Timing	Key results
Phase 1 to phase 2											
NCT05203224	Recruiting	EXTEND- IA DNase	Phase 2	300	2022	NETs	Dornase alfa (DNase)	Yes (IVT/ EVT)	NET degra- dation	Hyper- acute	N/A
NCT05880524	Not recruit- ing	ReSCInD	Phase 2	36	2024			Yes (STC)			
NCT02002390	Completed	N/A	Phase 2	22	2012	Lympho- cytes	Fingolimod	Yes/ IVT	Binds S1P1/ S1P3-5; sequesters lympho- cytes in lymph nodes	Hyper- acute/ acute	Reduced circu- lating lympho- cytes; smaller infarcts; less hemorrhage; improved early deficits and 90-d outcomes
NCT04629872	Unknown status	N/A	Phase 2	30	2020			Yes (EVT)		Hyper- acute	N/A
NCT04675762	Recruiting	N/A	Phase 2	118	2021			Yes (EVT/ IVT)		Hyper- acute	N/A
NCT04734548	Completed	APRIL	Phase 1 Phase 2	151	2020	TLR4	ApTOLL	Yes (EVT±IVT)	TLR4- inhibiting aptamer	Hyper- acute	Safe; potential reduction in mortality and disability at 90 d; reduced infarct volume, progression, and edema
NCT06380699	Completed	N/A	Phase 1	56	2023	BR-2	QHRD106	Healthy subjects	Long-acting tissue- kallikrein analogue; activates B2R, pro- moting vasodilation; reduces HMGB1	Hyper- acute/ acute	Safe, effective, long-acting kal- likrein drug
NCT06388772				74	2021			Healthy subjects			

The titles of all clinical trials are provided in Table S1. B2R indicates bradykinin receptor B2; EVT, endovascular treatment; GPIIb/IIIa, glycoprotein IIb/IIIa; GPVI, glycoprotein VI; HIF-1 α , hypoxia-inducible factor-1 α ; HMGB1, high mobility group box 1; IVT, intravenous thrombolysis; MMP, matrix metalloproteinase; MW, molecular weight; N/A, not available/not applicable; NET, extracellular trap; ROS, reactive oxygen species; STC, standard of care; and TLR, toll-like receptor.

Inflammation/Neuroinflammation

Several early phase and late-phase trials aim to modulate the inflammatory response as adjunctive therapy to reperfusion, seeking to mitigate detrimental cytokine release, microglial overactivation, and matrix degradation while preserving endogenous repair mechanisms.

Matrix Metalloproteinase Inhibition and Microglia Modulation

MMP (matrix metalloproteinase)-9 plays a pivotal role in BBB disruption, extracellular matrix degradation, and hemorrhagic transformation after reperfusion. Minocycline, a tetracycline derivative with anti-inflammatory, anti-apoptotic, and microglia-modulating properties, inhibits MMP-9 and downregulates HIF-1 α (hypoxia-inducible factor-1 α). Embolic diabetic rat models demonstrated significant reductions in infarct size.⁴⁸ The recently completed EMPHASIS (Efficacy and Safety of Minocycline in Patients With Moderate to Severe Acute Ischemic

Stroke; NCT05836740) phase III trial used minocycline during 4.5 days within 72 hours after AIS and showed a modest but significant 5.2% increase in patients achieving a modified Rankin Scale score <2 in the minocycline arm.⁴⁹ The real-world MIST-A/B (Effects of Minocycline on Patients With Ischemic Stroke Undergoing Intravenous Thrombectomy and Minocycline for Acute Ischemic Stroke Undergoing Endovascular Treatment due to Basilar Artery Occlusion; NCT05487417/NCT05512910) extensions will clarify whether minocycline can become the first clinically validated anti-inflammatory agent for AIS 90-day outcome improvement.

Cytokine Modulation

Tocilizumab, a humanized monoclonal antibody targeting the IL-6 receptor (membrane-bound and soluble), dampens downstream inflammatory amplification and endothelial activation. Aged and young rodent reperfusion models demonstrated both reduced infarct volume and

expression of endothelial adhesion molecules.⁵⁰ These observations motivated the phase II/III IRIS trial (Effect of Interleukin-6 Receptor Inhibition in Patients With Ischemic Stroke Undergoing Endovascular Treatment; NCT06238024), which reported that a single 240-mg intravenous dose administered alongside EVT within 24 hours after stroke onset resulted in a significant reduction in infarct core volume compared with EVT alone of 0.41 mL and fewer symptomatic intracranial hemorrhages (6% versus 12%).⁵¹ This aligns with early IL-6 elevation in patients with AIS and supports IL-6 pathway inhibition as a biologically coherent target although future trials are necessary to confirm the tocilizumab long-term benefit on functional outcome.

Leukocyte-Targeting Approaches

Neutrophils and macrophages are among the first leukocytes infiltrating the brain parenchyma after experimental stroke (Figure 1).^{8,9,20} These kinetics underscore the importance of timing when designing leukocyte-targeted therapies. Thus, leukocyte-modulation therapies aim to reduce secondary tissue damage (from inflammatory infiltration, microvascular occlusion, and endothelial disruption) and widen the therapeutic window of reperfusion therapies (Figure 2).

TLR4

ApTOLL is a single-stranded DNA aptamer that antagonizes TLR4, thereby attenuating innate immune responses.⁵² Rodent stroke models confirmed reduced infarct size, suppressed microglial activation, and improved survival.⁵³ NHP studies excluded secondary drug interactions or renal/hepatic toxicity.⁵⁴ ApTOLL has been evaluated in a phase Ib/IIa randomized clinical trial (NCT04734548) and administered before EVT within 6 hours of stroke onset to EVT candidates and was associated with a potential reduction in mortality and disability at 90 days, infarct volume, growth, and lower-grade edema.⁵² The inclusion of NHPs strengthens its translational relevance and addresses limitations of single-species preclinical strategies. ApTOLL acts in the systemic immune compartment but targets intracranial inflammatory signaling at sites where danger-associated molecular pattern-activated leukocytes accumulate after ischemia. Early administration alongside reperfusion therapy allows synergistic modulation of inflammation without impairing intravenous thrombolysis (IVT)/EVT efficacy, overcoming the safety and interaction issues that limited previous anti-inflammatory approaches, particularly in patients with incomplete recanalization and fast progressors.⁵²

Sphingosine-1-Phosphate

Fingolimod, a sphingosine-1-phosphate receptor modulator, prevents lymphocyte migration out of lymph nodes,

thereby reducing their circulation and infiltration into the ischemic brain. A preclinical meta-analysis (Stroke Preclinical Assessment Network), incorporating comorbidities and aged MCAO models, suggested that its effect varies and may not improve outcomes.⁵⁵ A pilot phase II trial suggested favorable outcomes in patients with AIS, with smaller infarct sizes and reduced rates of bleeding.⁵⁶ Ongoing phase II trial (NCT04675762) is investigating how fingolimod associated with EVT/IVT reperfusion improves clinical and radiological outcomes. Though fingolimod exemplifies selective lymphocyte modulation with favorable results in humans, preclinical data suggest that fingolimod's efficacy may be highly context dependent (sex, age, and comorbidities). Such limitations may contribute to failure in translation.

Thromboinflammation

Current trials aim to disrupt platelet-leukocyte and neutrophil-driven mechanisms that impair tissue perfusion despite large vessel recanalization.

Glycoprotein IIb/IIIa Complex

Tirofiban, a GPIIb/IIIa (glycoprotein IIb/IIIa) antagonist, prevents platelet aggregation and platelet-leukocyte complex formation. Preclinical stroke models have shown that tirofiban attenuates infarct volume, reduces oxidative stress, and modulates microglial activation while enhancing reperfusion.⁵⁷ Unlike abciximab (Table 1), tirofiban does not cross-react with receptors on other cells and exhibits reversible inhibition, potentially minimizing hemorrhagic risk.⁵⁸ A recent trial demonstrated that low-dose tirofiban administered after IVT was associated with improved neurological outcomes, supporting its potential as an adjunctive therapy to enhance reperfusion without significantly increasing bleeding risk.⁵⁹ Larger phase III/IV studies, including ADJUVANT (Intravenous Tirofiban Versus Alteplase Before Mechanical Thrombectomy in Stroke; NCT05728333) and ATTIS (Early Administration of Tirofiban in Patients Treated With Tenecteplase for Acute Ischemic Stroke; NCT05604638) in China, are underway to assess its efficacy in IVT-/EVT-treated patients. To date, tirofiban selectivity has demonstrated that precise platelet inhibition can enhance reperfusion without sacrificing safety.

Glycoprotein VI Receptor

Glenzocimab, a humanized Fab fragment targeting GPVI (glycoprotein VI) receptor, reduces platelet adhesion to collagen and fibrin without completely blocking aggregation.⁶⁰ GPVI depletion in an MCAO model significantly decreased infarct volume and neuronal death.⁶¹ NHP studies confirmed rapid inhibition of platelet aggregation without increased bleeding risk.⁶⁰ ACTISAVE (Acute Ischemic Stroke Study Evaluating Glenzocimab Used as Add-On Therapy Versus Placebo;

NCT05070260), GREEN (Glenzocimab for Reperfusion in the Setting of Endovascular Therapy for Brain Infarction; NCT05559398), and GALICE (Glenzocimab in Anterior Stroke With Large Ischemic Core Eligible for Endovascular Therapy; NCT06437431) are evaluating, in phase II-III studies, glenzocimab safety and efficacy in improving functional recovery. Partial GPVI inhibition emerges as a next-generation antiplatelet strategy for microvascular rescue. Glenzocimab acts intravascularly, selectively moderating platelet-collagen interactions at the lesion site. Its early administration (onset ≤ 4.5 hours) alongside IVT/EVT allows modulation of microvascular platelet adhesion precisely when distal embolization and microthrombi form, providing synergistic protection of intracranial microcirculatory flow without compromising reperfusion therapies or increasing bleeding risk. Nevertheless, the efficacy of glenzocimab may depend on patient selection, particularly those with no-reflow driven by platelet-rich microthrombi.

Neutrophil Extracellular Traps

Dornase alfa is a DNase I that degrades extracellular DNA scaffolds within neutrophil extracellular traps (NETs), which promote microvascular obstruction during ischemic events, representing a novel and mechanistically precise route to enhance distal reperfusion. Strong preclinical evidence across multiple rodent models has shown that NET targeting reduces infarct size and improves microcirculatory flow.^{62,63} EXTEND-IA DNase (Improving Early Reperfusion With Adjuvant Dornase Alfa in Large Vessel Ischemic Stroke; NCT05203224) and ReSCInD (Reduction of Systemic Inflammation After Ischemic Stroke by Intravenous DNase Administration; NCT05880524) trials are assessing DNase I as an adjuvant therapy to thrombolytics in anterior-circulation occlusions. Solid preclinical data, combined with an approach targeting complementary mechanisms, thrombolysis/EVT alongside NET degradation, hold promise for phase III trials. While improving microvessel reperfusion is a universal target, not all patients with large vessel occlusion stroke may respond to DNase therapy because clot composition is highly heterogeneous across stroke etiologies, and NET content varies significantly (Figure S1).

Kallikrein-Bradykinin System Modulation

The kallikrein-kinin system is an indispensable inflammatory modification system in vivo (Figures 1 and 2). The kallikrein-bradykinin pathway remains a promising but complex target in AIS: modulation can influence BBB permeability and vasogenic edema, yet balancing neuroprotection versus potential adverse effects has proven challenging.⁸

Low-Molecular-Weight Kininogen

HUK (human urinary-kallidinogenase) cleaves low-molecular-weight kininogen, enhancing tissue-kallikrein

activity and promoting vasodilation and microcirculatory perfusion.⁶⁴ Earlier trials with human urinary-kallidinogenase (NCT04102956/NCT02562183) supported these findings, reporting improvement in neurological function and cerebral perfusion.⁶⁵ However, a meta-analysis reviewing 9 studies in mainly MCAO/reperfusion rodent models, including hypertensive mice, demonstrated that human urinary-kallidinogenase resulted in a modest but significant 4.52% reduction in infarct sizes.⁶⁴ Limited preclinical benefit and indirect clinical and geographically restricted evidence may suggest a minimal gain in the clinical scenario. Ongoing phase 4 NCT06848894 study will assess the efficacy and safety of human urinary-kallidinogenase treatment together with EVT/IVT in >1200 patients.⁶⁶

Tissue Kallikrein

DM199 is a recombinant form of human-tissue kallikrein that enhances endogenous bradykinin production and nitric oxide release, aiming to restore microvascular function and ischemic penumbra perfusion.⁶⁷ In rodent models, tissue-kallikrein expression diminished salt-induced stroke mortality and vascular dysfunction, and reduced infarct volume and ischemia-induced locomotor deficit scores in MCAO models.^{68,69} In addition, plasma levels of tissue kallikrein were negatively associated with the risk of stroke in China.⁷⁰ Building up on this evidence, in the ReMEDy2 phase II/III trial (Treatment of Acute Ischemic Stroke; NCT05065216), DM199 efficacy on neurological recovery is currently under study. DM199 administration in the early postreperfusion window targets microvascular failure and capillary no-reflow, offering a physiological vasoprotective approach better tolerated than earlier systemic vasodilators. Although preclinical evidence from comorbidity-including models is robust, ReMEDy2 is being performed in nonrecanalized patients, which may limit DM199 efficacy.

QHRD106 is a PEG (polyethylene glycol)-modified, long-acting TK, designed to activate the bradykinin B2 receptor pathway.⁷¹ Its efficacy data currently rely on a single preclinical model. It should be viewed as a mechanistically attractive but early stage candidate.

Reactive Oxygen Species Inhibition

During ischemia reperfusion, reactive oxygen species production promotes lipid peroxidation, protein nitration, mitochondrial dysfunction, and activation of redox-sensitive enzymes, such as MMP-9, collectively compromising BBB integrity, exacerbating vasogenic edema, and potentiating hemorrhagic transformation.⁷² While antioxidant treatments have shown promise in laboratory studies, translating them into effective human therapies has been difficult.

In a recent study using human cerebral organoids, edaravone reduced cell death and apoptosis in a

dose-dependent manner, confirming its antioxidant and neuroprotective effects in human neural tissue.⁷³ A recent phase II randomized trial using intravenous edaravone dextroboenol in successfully recanalized patients after EVT was safe and well tolerated but did not improve 90-day functional outcomes.⁷⁴ By contrast, TASTE-SL (Phase III Clinical Trial of Y-2 Sublingual Tablets in the Treatment of Acute Ischemic Stroke; NCT04950920) administered sublingual Y-2 (edaravone dextroboenol) for 14 days after AIS onset, increasing independence by 9.7% (90-day modified Rankin Scale score ≤ 1), indicating a clinically meaningful neuroprotective effect.⁷⁵ Generalization must be cautious as the cohort included predominantly young patients with small-vessel disease and large-artery atherosclerosis. Moreover, because edaravone dextroboenol represents dual-mechanism modulation of oxidative stress and neuroinflammation where formulation and administration route may be critical to achieving consistent neuroprotective efficacy, further validation trials considering more heterogeneous cohorts and from different backgrounds are needed.

CONCLUSIONS AND FUTURE DIRECTIONS

Despite decades of effort, neuroprotective and anti-inflammatory AIS therapies have repeatedly failed to translate. Most therapeutic agents targeted biologically relevant pathways but may have been applied at sub-optimal times, in the wrong compartment, or without considering the physiological context created by recanalization. In addition, preclinical models, usually young and comorbidity-free, do not reflect the clinical reality of delayed reperfusion, human clot biology, or comorbid immune dysfunction.

Data emerging from the postthrombectomy era demonstrate that the critical therapeutic window extends beyond macrovascular recanalization and lies within the intracranial and microvascular phases of injury. Several late-phase trials, supported by promising results from earlier studies, are now addressing some of these mechanisms directly. Studies such as ADJUVANT (platelet/platelet-leukocyte interactions) and UNITE (Efficacy and Safety of Human Urinary Kallidinogenase for Acute Ischemic Stroke Patients Receiving Reperfusion Treatment; kallikrein pathway; NCT06848894) are nearing completion and may clarify whether mechanism-aligned therapies can improve functional outcomes when deployed in synchrony with reperfusion, as already demonstrated in EMPHASIS (microglial modulation).

In parallel, emergent agents acting on NET formation (DNase I), GPVI inhibition (glenzocimab), TLR4 antagonism (ApTOLL), IL-6 inhibition (tocilizumab), and kallikrein pathway modulators (DM199) illustrate a new

generation of target-precise, reperfusion-synchronized interventions. These strategies operate on top of recanalization, protecting the microcirculation, stabilizing the endothelial barrier, and attenuating deleterious thromboinflammation that occurs after recanalization.

Advancing the field will likely require embedding therapies into the reperfusion workflow (pre-EVT, intraprocedural, and early postrecanalization), guided by mechanistic-specific biomarkers, intracranial sampling, and imaging of microvascular patency. Preclinical pipelines must evolve toward humanized models, comorbidity-inclusive, reperfusion-based, which integrate IVT/EVT, delayed microvascular failure, accounting for the heterogeneity of human thrombi to best reproduce the thromboinflammatory environment.

Ultimately, precision stroke therapy will require modulation of grounded targets, such as NETs, GPVI, complement-kallikrein crosstalk, and innate immune receptors aligned with reperfusion, to timely target the right mechanism in the right physiological compartment.

ARTICLE INFORMATION

Received November 20, 2025; final revision received April 16, 2026; accepted May 11, 2026.

Affiliations

ICBAS-Instituto de Ciências Biomédicas Abel Salazar (A.D., A.M., J.M., L.F.M.) and i3S-Instituto de Investigação e Inovação em Saúde (A.D., A.M., L.F.M.), University of Porto, Portugal. Department of Neuroradiology (L.I.) and Department of Neurology (J.M., L.F.M.), ULSSA-Unidade Local de Saúde de Santo António, Porto, Portugal. Department of Neurology, University Hospital Zürich and UZH-University of Zürich, Switzerland (S.W.).

Sources of Funding

Dr Wegener was supported by the Swiss National Science Foundation (grant 3200-0-239964) and the Swiss Heart Foundation.

Disclosures

Dr Wegener received research funds by the Swiss National Science Foundation, the University of Zürich (UZH) Clinical Research Priority Program Stroke, the Hartmann Müller Foundation, the Koetser Foundation, the Philas Foundation, and the Swiss Heart Foundation; reports speaker honoraria from Springer, Advisis AG, Teva Pharma, Boehringer Ingelheim, Desitin Pharma, Lundbeck, AstraZeneca, and Forum für Medizinische Fortbildung (FoMF); and received consultancy fee from Teva Pharma outside the submitted work. The other authors report no conflicts.

Supplemental Material

Table S1
Figures S1–S2

REFERENCES

- Hirsch S, Kostev K, Tanislav C, Hamed A. Acute ischemic stroke treatment in Germany (2015–2023): nationwide trends in thrombolysis and thrombectomy by age and sex. *Brain Sci*. 2025;15:832. doi: 10.3390/brainsci15080832
- Stoll G, Nieswandt B, Schuhmann MK. Ischemia/reperfusion injury in acute human and experimental stroke: focus on thrombo-inflammatory mechanisms and treatments. *Neurol Res Pract*. 2024;6:57. doi: 10.1186/s42466-024-00355-y
- Dirnagl U, Iadecola C, Moskowitz MA. Pathobiology of ischaemic stroke: an integrated view. *Trends Neurosci*. 1999;22:391–397. doi: 10.1016/s0166-2236(99)01401-0

4. Koçkar IN, Koyuncuoğlu HR. Pathophysiology of parenchymal injury in ischemic stroke. *AJNN*. 2024;1:5–10. doi: 10.51271/ajnn-0002
5. Chamorro A, Meisel A, Planas AM, Urra X, van de Beek D, Veltkamp R. The immunology of acute stroke. *Nat Rev Neurol*. 2012;8:401–410. doi: 10.1038/nrneurol.2012.98
6. Kreisl WC, Kim MJ, Coughlin JM, Henter ID, Owen DR, Innis RB. PET imaging of neuroinflammation in neurological disorders. *Lancet Neurol*. 2020;19:940–950. doi: 10.1016/S1474-4422(20)30346-X
7. Shen X-Y, Gao Z-K, Han Y, Yuan M, Guo Y-S, Bi X. Activation and role of astrocytes in ischemic stroke. *Front Cell Neurosci*. 2021;15:755955. doi: 10.3389/fncel.2021.755955
8. Szeponowski RD, Hauptelshofer S, Vonhof SE, Frank B, Kleinschnitz C, Casas AI. Thromboinflammatory challenges in stroke pathophysiology. *Semin Immunopathol*. 2023;45:389–410. doi: 10.1007/s00281-023-00994-4
9. Stoll G, Schuhmann MK, Nieswandt B, Kollikowski AM, Pham M. An intravascular perspective on hyper-acute neutrophil, T-cell and platelet responses: similarities between human and experimental stroke. *J Cereb Blood Flow Metab*. 2022;42:1561–1567. doi: 10.1177/0271678X221105764
10. Offner H, Subramanian S, Parker SM, Afentoulis ME, Vandenbark AA, Hurn PD. Experimental stroke induces massive, rapid activation of the peripheral immune system. *J Cereb Blood Flow Metab*. 2006;26:654–665. doi: 10.1038/sj.jcbfm.9600217
11. Denes A, McColl BW, Leow-Dyke SF, Chapman KZ, Humphreys NE, Grecnis RK, Allan SM, Rothwell NJ. Experimental stroke-induced changes in the bone marrow reveal complex regulation of leukocyte responses. *J Cereb Blood Flow Metab*. 2011;31:1036–1050. doi: 10.1038/jcbfm.2010.198
12. Liu ZJ, Chen C, Li FW, Shen JM, Yang YY, Leak RK, Ji XM, Du HS, Hu XM. Splenic responses in ischemic stroke: new insights into stroke pathology. *CNS Neurosci Ther*. 2015;21:320–326. doi: 10.1111/cns.12361
13. Iadecola C, Buckwalter MS, Anrather J. Immune responses to stroke: mechanisms, modulation, and therapeutic potential. *J Clin Invest*. 2020;130:2777–2788. doi: 10.1172/JCI135530
14. Kollikowski AM, Schuhmann MK, Nieswandt B, Müllges W, Stoll G, Pham M. Local leukocyte invasion during hyperacute human ischemic stroke. *Ann Neurol*. 2020;87:466–479. doi: 10.1002/ana.25665
15. Kollikowski AM, Pham M, März AG, Papp L, Nieswandt B, Stoll G, Schuhmann MK. Platelet activation and chemokine release are related to local neutrophil-dominant inflammation during hyperacute human stroke. *Transl Stroke Res*. 2022;13:364–369. doi: 10.1007/s12975-021-00938-w
16. El Amki M, Glück C, Binder N, Middleham W, Wyss MT, Weiss T, Meister H, Luft A, Weller M, Weber B, et al. Neutrophils obstructing brain capillaries are a major cause of no-reflow in ischemic stroke. *Cell Rep*. 2020;33:108260. doi: 10.1016/j.celrep.2020.108260
17. Baek JH, Kim BM, Heo JH, Nam HS, Kim YD, Park H, Bang OY, Yoo J, Kim DJ, Jeon P, et al. Number of stent retriever passes associated with futile recanalization in acute stroke. *Stroke*. 2018;49:2088–2095. doi: 10.1161/STROKEAHA.118.021320
18. Deng G, Xiao J, Yu H, Chen M, Shang K, Qin C, Tian DS. Predictors of futile recanalization after endovascular treatment in acute ischemic stroke: a meta-analysis. *J Neurointerv Surg*. 2022;14:881–885. doi: 10.1136/neurintsurg-2021-017963
19. Lee DH, Sung JH, Kim SU, Yi HJ, Hong JT, Lee SW. Effective use of balloon guide catheters in reducing incidence of mechanical thrombectomy related distal embolization. *Acta Neurochir (Wien)*. 2017;159:1671–1677. doi: 10.1007/s00701-017-3256-3
20. Stoll G, Nieswandt B. Thrombo-inflammation in acute ischaemic stroke - implications for treatment. *Nat Rev Neurol*. 2019;15:473–481. doi: 10.1038/s41582-019-0221-1
21. Holloway PM, Gavins FN. Modeling ischemic stroke in vitro: status quo and future perspectives. *Stroke*. 2016;47:561–569. doi: 10.1161/STROKEAHA.115.011932
22. Van Breedam E, Ponsaerts P. Promising strategies for the development of advanced in vitro models with high predictive power in ischaemic stroke research. *Int J Mol Sci*. 2022;23:7140. doi: 10.3390/ijms23137140
23. Babu M, Singh N, Datta A. In vitro oxygen glucose deprivation model of ischemic stroke: a proteomics-driven systems biological perspective. *Mol Neurobiol*. 2022;59:2363–2377. doi: 10.1007/s12035-022-02745-2
24. Wang R, Li L, Wang B. Poncirin ameliorates oxygen glucose deprivation/perfusion injury in cortical neurons via inhibiting NOX4-mediated NLRP3 inflammasome activation. *Int Immunopharmacol*. 2022;102:107210. doi: 10.1016/j.intimp.2020.107210
25. He Z, Dony L, Fleck JS, Szalata A, Li KX, Slišković I, Lin H-C, Santel M, Atamian A, Quadrato G, et al; Human Cell Atlas Organoid Biological Network. An integrated transcriptomic cell atlas of human neural organoids. *Nature*. 2024;635:690–698. doi: 10.1038/s41586-024-08172-8
26. De Paola M, Pischiutta F, Comolli D, Mariani A, Kelk J, Lisi I, Cericov M, Fumagalli S, Forloni G, Zanier ER. Neural cortical organoids from self-assembling human iPSC as a model to investigate neurotoxicity in brain ischemia. *J Cereb Blood Flow Metab*. 2023;43:680–693. doi: 10.1177/0271678X231152023
27. Boerstler T, Kachkin D, Gerasimova E, Zagha N, Furlanetto F, Nayeibzade N, Zappia L, Boisvert M, Farrell M, Floetz S, et al. Deciphering brain organoid heterogeneity by identifying key quality determinants. *Commun Biol*. 2025;8:1412. doi: 10.1038/s42003-025-08855-6
28. Ingber DE. Human organs-on-chips for disease modelling, drug development and personalized medicine. *Nat Rev Genet*. 2022;23:467–491. doi: 10.1038/s41576-022-00466-9
29. Lyu Z, Park J, Kim KM, Jin HJ, Wu H, Rajadas J, Kim DH, Steinberg GK, Lee W. A neurovascular-unit-on-a-chip for the evaluation of the restorative potential of stem cell therapies for ischaemic stroke. *Nat Biomed Eng*. 2021;5:847–863. doi: 10.1038/s41551-021-00744-7
30. Li Z, Yu D, Zhou C, Wang F, Lu K, Liu Y, Xu J, Xuan L, Wang X. Engineering vascularised organoid-on-a-chip: strategies, advances and future perspectives. *Biomater Transl*. 2024;5:21–32. doi: 10.12336/biomatertransl.2024.01.003
31. Kaiser EE, West FD. Large animal ischemic stroke models: replicating human stroke pathophysiology. *Neural Regen Res*. 2020;15:1377–1387. doi: 10.4103/1673-5374.274324
32. Taha A, Bobi J, Dammers R, Dijkhuizen RM, Dreyer AY, van Es ACGM, Ferrara F, Gounis MJ, Nitzsche B, Platt S, et al. Comparison of large animal models for acute ischemic stroke: which model to use? *Stroke*. 2022;53:1411–1422. doi: 10.1161/strokeaha.121.036050
33. Sommer CJ. Ischemic stroke: experimental models and reality. *Acta Neuropathol*. 2017;133:245–261. doi: 10.1007/s00401-017-1667-0
34. Singh D, Wasan H, Reeta KH. Preclinical stroke research and translational failure: a bird's eye view on preventable variables. *Cell Mol Neurobiol*. 2022;42:2003–2017. doi: 10.1007/s10571-021-01083-6
35. Dias A, Silva L, Moura J, Gabriel D, Maia LF. Fluid biomarkers in stroke: from animal models to clinical care. *Acta Neurol Scand*. 2022;146:332–347. doi: 10.1111/ane.13668
36. Fluri F, Schuhmann MK, Kleinschnitz C. Animal models of ischemic stroke and their application in clinical research. *Drug Des Devel Ther*. 2015;9:3445–3454. doi: 10.2147/DDDT.S56071
37. Cotrina ML, Lou N, Tome-Garcia J, Goldman J, Nedergaard M. Direct comparison of microglial dynamics and inflammatory profile in photothrombotic and arterial occlusion evoked stroke. *Neuroscience*. 2017;343:483–494. doi: 10.1016/j.neuroscience.2016.12.012
38. Sharp FR, Jickling GC. Modeling immunity and inflammation in stroke: differences between rodents and humans? *Stroke*. 2014;45:e179–e180. doi: 10.1161/STROKEAHA.114.005639
39. Lin S, Lin Y, Nery JR, Urich MA, Breschi A, Davis CA, Dobin A, Zaleski C, Beer MA, Chapman WC, et al. Comparison of the transcriptional landscapes between human and mouse tissues. *Proc Natl Acad Sci U S A*. 2014;111:17224–17229. doi: 10.1073/pnas.1413624111
40. Koo BS, Lee DH, Kang P, Jeong KJ, Lee S, Kim K, Lee Y, Huh JW, Kim YH, Park SJ, et al. Reference values of hematological and biochemical parameters in young-adult cynomolgus monkey (Macaca fascicularis) and rhesus monkey (Macaca mulatta) anesthetized with ketamine hydrochloride. *Lab Anim Res*. 2019;35:7. doi: 10.1186/s42826-019-0006-0
41. Merck & Co, Inc. Hematology (Complete Blood Count) Reference Ranges. In: *Merck Veterinary Manual*. 2025. Accessed November 19, 2025. <https://www.merckvetmanual.com/multimedia/table/hematology-complete-blood-count-reference-ranges>
42. Bjornson-Hooper ZB, Fragiadakis GK, Spitzer MH, Chen H, Madhiredy D, Hu K, Lundsten K, McIlwain DR, Nolan GP. A comprehensive atlas of immunological differences between humans, mice, and non-human primates. *Front Immunol*. 2022;13:867015. doi: 10.3389/fimmu.2022.867015
43. Mosneag IE, Flaherty SM, Wykes RC, Allan SM. Stroke and translational research - review of experimental models with a focus on awake ischaemic induction and anaesthesia. *Neuroscience*. 2024;550:89–101. doi: 10.1016/j.neuroscience.2023.11.034
44. Lind BL, Kucharz K, Cai C. In vivo two-photon microscopy for studies of neurovascular coupling in rodents: a beginner's guide. *Neurophotonics*. 2025;12:S22808. doi: 10.1117/1.NPh.12.S22808
45. Veltkamp R, Gill D. Clinical trials of immunomodulation in ischemic stroke. *Neurotherapeutics*. 2016;13:791–800. doi: 10.1007/s13311-016-0458-y
46. Candelario-Jalil E, Paul S. Impact of aging and comorbidities on ischemic stroke outcomes in preclinical animal models: a translational perspective. *Exp Neurol*. 2021;335:113494. doi: 10.1016/j.expneurol.2020.113494

47. McCann SK, Lawrence CB. Comorbidity and age in the modelling of stroke: are we still failing to consider the characteristics of stroke patients? *BMJ Open Sci.* 2020;4:e100013. doi: 10.1136/bmjos-2019-100013
48. Fan X, Lo EH, Wang X. Effects of minocycline plus tissue plasminogen activator combination therapy after focal embolic stroke in type 1 diabetic rats. *Stroke.* 2013;44:745–752. doi: 10.1161/STROKEAHA.111.000309
49. Lu Y, Guan L, Wu J, Yang Q, Zhang M, Zhou D, Yang H, Pan Y, Wang L, Qiu B, et al; EMPHASIS Investigators. Efficacy and safety of minocycline in patients with acute ischaemic stroke (EMPHASIS): a multicentre, double-blind, randomised controlled trial. *Lancet.* 2026;407:679–688. doi: 10.1016/S0140-6736(25)01862-8
50. Hudobenko J, Chauhan A, McCullough L. Abstract 128: amelioration of ischemic stroke damage through inhibition of interleukin-6 signaling with tocilizumab requires sex specific dosing. *Stroke.* 2019;50:A128–A128. doi: 10.1161/str.50.suppl_1.128
51. Chu X, Lan J, Ma Z, Yin Y, Gu H, Baron JC, Liesz A, Wen C, Liu Y, Sun J, et al; IRIS Investigators. Effect of IL-6 receptor inhibition on infarct volume after endovascular treatment for ischaemic stroke: a phase 2, randomised, placebo-controlled trial. *EBioMedicine.* 2026;125:106167. doi: 10.1016/j.ebiom.2026.106167
52. Hernández-Pérez M, Valls-Carbó A, Hernández-Jiménez M, Molina C, Vivas Mora J, Castellanos M, García-Madróna S, Tembl JI, Moniche F, Terceño M, et al. Changes in infarct volume and cerebral edema after mechanical thrombectomy in patients with stroke randomized to ApTOLL or placebo: a secondary analysis of APRIL trial. *Stroke.* 2025;56:2422–2430. doi: 10.1161/STROKEAHA.124.048917
53. Fernández G, Moraga A, Cuartero MI, García-Culebras A, Peña-Martínez C, Pradillo JM, Hernández-Jiménez M, Sacristán S, Ayuso MI, Gonzalo-Gobernado R, et al. TLR4-binding DNA aptamers show a protective effect against acute stroke in animal models. *Mol Ther.* 2018;26:2047–2059. doi: 10.1016/j.jymthe.2018.05.019
54. Hernández-Jiménez M, Martín-Vilchez S, Ochoa D, Mejía-Abril G, Román M, Camargo-Mamani P, Luquero-Bueno S, Jilma B, Moro MA, Fernández G, et al. First-in-human phase I clinical trial of a TLR4-binding DNA aptamer, ApTOLL: safety and pharmacokinetics in healthy volunteers. *Mol Ther Nucleic Acids.* 2022;28:124–135. doi: 10.1016/j.omtn.2022.03.005
55. Boisserand LSB, Herman AL, Sanganahalli BG, Mihailovic J, Beatty HE, Johnson CW, Diaz S, DeLong JH, Velazquez S, Grutzendler J, et al; SPAN Investigators. Fingolimod as a potential cerebroprotectant results from the stroke preclinical assessment network. *Stroke.* 2025;56:3280–3293. doi: 10.1161/STROKEAHA.125.050903
56. Zhu Z, Fu Y, Tian D, Sun N, Han W, Chang G, Dong Y, Xu X, Liu Q, Huang D, et al. Combination of the immune modulator fingolimod with alteplase in acute ischemic stroke: a pilot trial. *Circulation.* 2015;132:1104–1112. doi: 10.1161/CIRCULATIONAHA.115.016371
57. Jiang Y, Huang W, Zhang Y, Ji Q. Tirofiban in acute ischemic stroke: mechanistic rationale, clinical advances, and emerging therapeutic strategies. *Drugs.* 2025;85:1269–1287. doi: 10.1007/s40265-025-02222-9
58. Topol EJ, Moliterno DJ, Herrmann HC, Powers ER, Grines CL, Cohen DJ, Cohen EA, Bertrand M, Neumann F-J, Stone GW, et al; TARGET Investigators. Do Tirofiban and ReoPro Give Similar Efficacy Trial. Comparison of two platelet glycoprotein IIb/IIIa inhibitors, tirofiban and abciximab, for the prevention of ischemic events with percutaneous coronary revascularization. *N Engl J Med.* 2001;344:1888–1894. doi: 10.1056/NEJM200106213442502
59. Wu C, Sun C, Wang L, Lian Y, Xie N, Huang S, Zhao W, Ren M, Wu D, Ding J, et al. Low-dose tirofiban treatment improves neurological deterioration outcome after intravenous thrombolysis. *Stroke.* 2019;50:3481–3487. doi: 10.1161/STROKEAHA.119.026240
60. Lebozec K, Jandrot-Perrus M, Avenard G, Favre-Bulle O, Billiard P. Design, development and characterization of ACT017, a humanized Fab that blocks platelet's glycoprotein VI function without causing bleeding risks. *Mabs.* 2017;9:945–958. doi: 10.1080/19420862.2017.1336592
61. Bieber M, Schuhmann MK, Kollikowski AM, Stegner D, Nieswandt B, Pham M, Stoll G. Targeting platelet glycoprotein VI attenuates progressive ischemic brain damage before recanalization during middle cerebral artery occlusion in mice. *Exp Neurol.* 2021;344:113804. doi: 10.1016/j.expneurol.2021.113804
62. Denorme F, Portier I, Rustad JL, Cody MJ, de Araujo CV, Hoki C, Alexander MD, Grandhi R, Dyer MR, Neal MD, et al. Neutrophil extracellular traps regulate ischemic stroke brain injury. *J Clin Invest.* 2022;132:e154225. doi: 10.1172/JCI154225
63. Di G, Vázquez-Reyes S, Díaz B, Peña-Martínez C, García-Culebras A, Cuartero MI, Moraga A, Pradillo JM, Esposito E, Lo EH, et al. Daytime DNase-I administration protects mice from ischemic stroke without inducing bleeding or tPA-induced hemorrhagic transformation, even with aspirin pretreatment. *Stroke.* 2025;56:527–532. doi: 10.1161/STROKEAHA.124.049961
64. Zhang N, Meng F, Bao W, Li X, He F, Wang A, Xu Z. Efficacy of human urinary kallidinogenase in ischemic stroke in animal models: a meta-analysis. *Int J Clin Exp Med.* 2019;12:8670–8677.
65. Wang YX, Chen Y, Zhang CH, Li CH, Dong Z, Zhao SN, Wang Z, Zhang FF, Tong XG, Wang JH, et al. Study on the effect of urinary kallidinogenase after thrombolytic treatment for acute cerebral infarction. *Eur Rev Med Pharmacol Sci.* 2015;19:1009–1012.
66. Wisniewski P, Gangnus T, Burckhardt BB. Recent advances in the discovery and development of drugs targeting the kallikrein-kinin system. *J Transl Med.* 2024;22:388. doi: 10.1186/s12967-024-05216-5
67. Campbell BC, Kasner SE, Lista AD, Volpi JJ, Kleinig TJ, Cordato D, Dewey HM, Choi PM, Garcia-Esperon C, Sahathevan R, et al; ReMEDy1 Investigators. Safety and tolerability of rinocalinase alfa (DM199) for acute ischemic stroke (ReMEDy1) [published online November 6, 2025]. *Int J Stroke.* 2025. doi: 10.1177/17474930251396480
68. Xia CF, Yin H, Borlongan CV, Chao L, Chao J. Kallikrein gene transfer protects against ischemic stroke by promoting glial cell migration and inhibiting apoptosis. *Hypertension.* 2004;43:452–459. doi: 10.1161/01.HYP.0000110905.29389.e5
69. Zhang JJ, Chao L, Chao J. Adenovirus-mediated kallikrein gene delivery reduces aortic thickening and stroke-induced death rate in Dahl salt-sensitive rats. *Stroke.* 1999;30:1925–31; discussion 1931. doi: 10.1161/01.str.30.9.1925
70. Zhang Q, Ding H, Yan J, Wang W, Ma A, Zhu Z, Cianflone K, Hu FB, Hui R, Wang DW. Plasma tissue kallikrein level is negatively associated with incident and recurrent stroke: a multicenter case-control study in China. *Ann Neurol.* 2011;70:265–273. doi: 10.1002/ana.22404
71. Xu SY, Jia JQ, Sun M, Bao XY, Xia SN, Shu S, Liu PY, Ji SL, Ye L, Cao X, et al. QHRD106 ameliorates ischemic stroke injury as a long-acting tissue kallikrein preparation. *iScience.* 2023;26:107268. doi: 10.1016/j.isci.2023.107268
72. Allen CL, Bayraktutan U. Oxidative stress and its role in the pathogenesis of ischaemic stroke. *Int J Stroke.* 2009;4:461–470. doi: 10.1111/j.1747-4949.2009.00387.x
73. Wang S-N, Wang Z, Wang X-Y, Zhang X-P, Xu T-Y, Miao C-Y. Humanized cerebral organoids-based ischemic stroke model for discovering of potential anti-stroke agents. *Acta Pharmacol Sin.* 2022;44:513–523. doi: 10.1038/s41401-022-00986-4
74. Chen HS, Zhao ZA, Shen XY, Qiu SQ, Cui Y, Qiu J, Li W, Zhang H, Chen WH, Wang LH, et al. Edaravone dextran for ischemic stroke with sufficient recanalization after thrombectomy: a randomized phase II trial. *Nat Commun.* 2025;16:2393. doi: 10.1038/s41467-025-57774-x
75. Fu Y, Wang A, Tang R, Li S, Tian X, Xia X, Ren J, Yang S, Chen R, Zhu S, et al. Sublingual edaravone dextran for the treatment of acute ischemic stroke: the TASTE-SL randomized clinical trial. *JAMA Neurol.* 2024;81:319–326. doi: 10.1001/jamaneurol.2023.5716